

Dark Matter Residuals as Structural Coherence Diagnostics in MAAT v1.6

A Conceptual Diagnostic Note on Galactic Rotation Residuals

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Abstract

Dark matter is normally introduced as an additional gravitating component required by galaxy rotation curves, lensing, structure formation, and the cosmic microwave background. This note does not attempt to replace that standard interpretation and does not argue against particle dark matter. Instead, it asks a narrower phenomenological question: how would MAAT v1.6 read the dark-matter signal as a structural residual?

In MAAT v1.6, complex systems are described as trajectories in structured search spaces. A residual is not merely an error term; it marks where the visible degrees of freedom fail to close the observed dynamical pattern. For galactic rotation curves, the effective residual

$$v_{\text{dark}}^2(r) = \max(v_{\text{obs}}^2(r) - v_{\text{bar}}^2(r), 0)$$

is therefore interpreted as the missing structural channel required to keep the observed gravitational trajectory coherent.

A synthetic rotation-curve benchmark is used to compute MAAT support fields $(H, B, S_{\text{eff}}, V)$, the v1.2.1 robustness closure R_{rob} , and a structural dark support $D_{\text{struct}}(r)$. The toy model yields an outer residual fraction 0.8802, an outer mean robustness 0.7806, and a peak structural dark support 0.3486 near 20.38 kpc. These numbers are not astronomical constraints and do not constitute evidence for a new dark-matter model. All numerical values reported here refer only to the synthetic benchmark and should not be interpreted as measurements of real galaxies. The contribution is limited but useful: the note turns a familiar rotation-curve residual into an explicit, reproducible structural diagnostic and identifies what would be required for a nontrivial test on real data.

1 Purpose and Guardrail

This is a phenomenological note. It is not a dark-matter detection paper, not a galaxy-fit paper, and not a replacement for cold dark matter.

The expected criticism is correct: the basic residual

$$v_{\text{obs}}^2 - v_{\text{bar}}^2$$

is not new. It is the standard dark-matter problem written in rotation-curve language. Therefore the present note should not be read as a discovery claim. It is a translation and sanity-check exercise: can the residual be expressed as a structural diagnostic without immediately overclaiming its microscopic meaning?

The purpose is conceptual and diagnostic:

Dark matter is treated as the gravitationally inferred structural residual required to keep the observed dynamical trajectory coherent after visible contributions are accounted for.

This statement is compatible with particle dark matter. If the residual is carried by a new particle sector, MAAT would interpret that sector as the physical carrier of a structural coherence channel. If some part of the residual arises from modified gravity, baryonic feedback, or projection effects, MAAT would instead interpret those mechanisms as alternative carriers of the same inferred residual channel.

Thus the paper does not answer the microscopic question

What particle is dark matter?

It asks the structural question

What role does the dark residual play in closing galactic dynamics?

2 Dark Matter as a Residual

For a circular orbit in a galaxy, the observed rotation curve is usually written schematically as

$$v_{\text{obs}}^2(r) = v_{\text{bar}}^2(r) + v_{\text{dark}}^2(r), \quad (1)$$

where v_{bar} denotes the visible baryonic contribution and v_{dark} denotes the effective dark contribution inferred from gravity.

The residual is therefore

$$v_{\text{dark}}^2(r) = \max[v_{\text{obs}}^2(r) - v_{\text{bar}}^2(r), 0]. \quad (2)$$

The corresponding residual fraction is

$$f_{\text{res}}(r) = \frac{v_{\text{dark}}^2(r)}{v_{\text{obs}}^2(r)}. \quad (3)$$

In standard phenomenology, this residual is interpreted as missing mass or an effective gravitational source. MAAT v1.6 adds a diagnostic layer: the residual is also a measure of where the visible state description fails to close the observed dynamical trajectory.

3 MAAT v1.6 Interpretation

MAAT v1.6 treats complex systems as trajectories through structured search spaces. The relevant question is not only whether a configuration exists, but whether it remains coherent, balanced, dynamically active, connected, and robust under the operators that test it.

For the present toy galaxy profile we use four supports:

- $H(r)$: reconstruction consistency after adding the residual channel.
- $B(r)$: radial force-balance smoothness of the observed trajectory.
- $S_{\text{eff}}(r)$: controlled transition activity as the residual emerges.
- $V(r)$: coupling between baryonic structure and residual-density shape.

These definitions are heuristic. They are not derived from a microscopic dark-sector action or from a relativistic modified-gravity theory. Their status in this note is only operational:

Support	Toy meaning	Caveat
H	Does the added residual reconstruct the observed curve?	Nearly tautological here, because the residual is defined from the reconstruction condition.
B	Is the radial trajectory dynamically smooth rather than violently oscillatory?	A smoothness proxy, not a full Jeans or disk-dynamics condition.
S_{eff}	Is the transition into the residual-dominated regime controlled?	Chosen as a bounded activity proxy, not uniquely derived.
V	Is the residual-density shape connected to the baryonic structure?	Only a shape-coupling proxy; not a baryonic feedback model.

The v1.2.1 robustness closure is retained:

$$R_{\text{resp}}(r) = [H(r)B(r)V(r)]^{1/3}, \quad (4)$$

$$R_{\text{rob}}(r) = \min \left[R_{\text{resp}}(r), (H(r)B(r)S_{\text{eff}}(r)V(r))^{1/4} \right]. \quad (5)$$

The structural dark support is then defined as

$$D_{\text{struct}}(r) = f_{\text{res}}(r) R_{\text{rob}}(r) V(r). \quad (6)$$

This quantity is deliberately not identical to the dark residual. A large residual alone is not enough. The residual receives structural support only when it is robustly connected to the coherent dynamical profile.

4 Toy Benchmark

The benchmark uses a synthetic galaxy profile. The observed curve is chosen to rise and approach a flat outer value,

$$v_{\text{obs}}(r) = v_{\text{flat}} \left(1 - e^{-r/r_{\text{turn}}} \right), \quad (7)$$

with $v_{\text{flat}} \simeq 205 \text{ km s}^{-1}$.

The baryonic contribution is an exponential-disk-like proxy plus a weak gas floor. The dark residual is then inferred from Eq. (2). No external galaxy data are used.

5 Why This Benchmark Is Weak

This first benchmark is intentionally simple, and that simplicity is also its main weakness. The observed curve is synthetic and chosen to become flat. The baryonic contribution is a smooth toy profile. The dark residual is then constructed by subtraction. Consequently, the experiment cannot show that MAAT predicts dark matter, fits galaxies, or improves on existing halo or modified-dynamics models.

In particular, $H \simeq 1$ should not be interpreted as a discovery. It only confirms that a residual defined by Eq. (2) reconstructs the toy curve by construction. The only mildly nontrivial quantities in this note are the balance, activity, connectedness, and robustness profiles built on top of the residual.

The scientific value is therefore narrow:

The note provides a vocabulary and a reproducible diagnostic scaffold, not a new empirical result about dark matter.

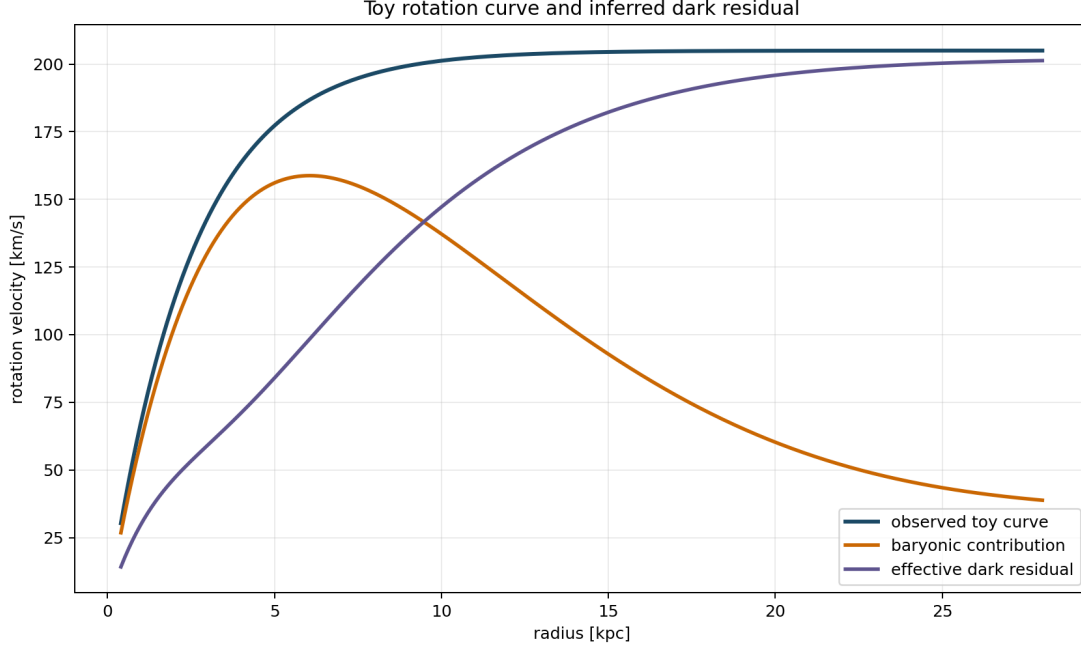


Figure 1: Synthetic rotation curve, baryonic contribution, and effective dark residual. The dark residual is the inferred channel needed to maintain the flat observed trajectory after the baryonic component declines.

6 Results

The benchmark gives the following summary.

Table 1: Toy dark-residual benchmark summary.

Quantity	Value
maximum observed velocity	$204.9972 \text{ km s}^{-1}$
outer mean residual fraction	0.8802
outer mean robustness R_{rob}	0.7806
transition peak structural dark support	0.3486
radius of peak structural dark support	20.3782 kpc
mean H	1.0000
mean B	0.9060
mean S_{eff}	0.7506
mean V	0.5919
mean R_{rob}	0.7611

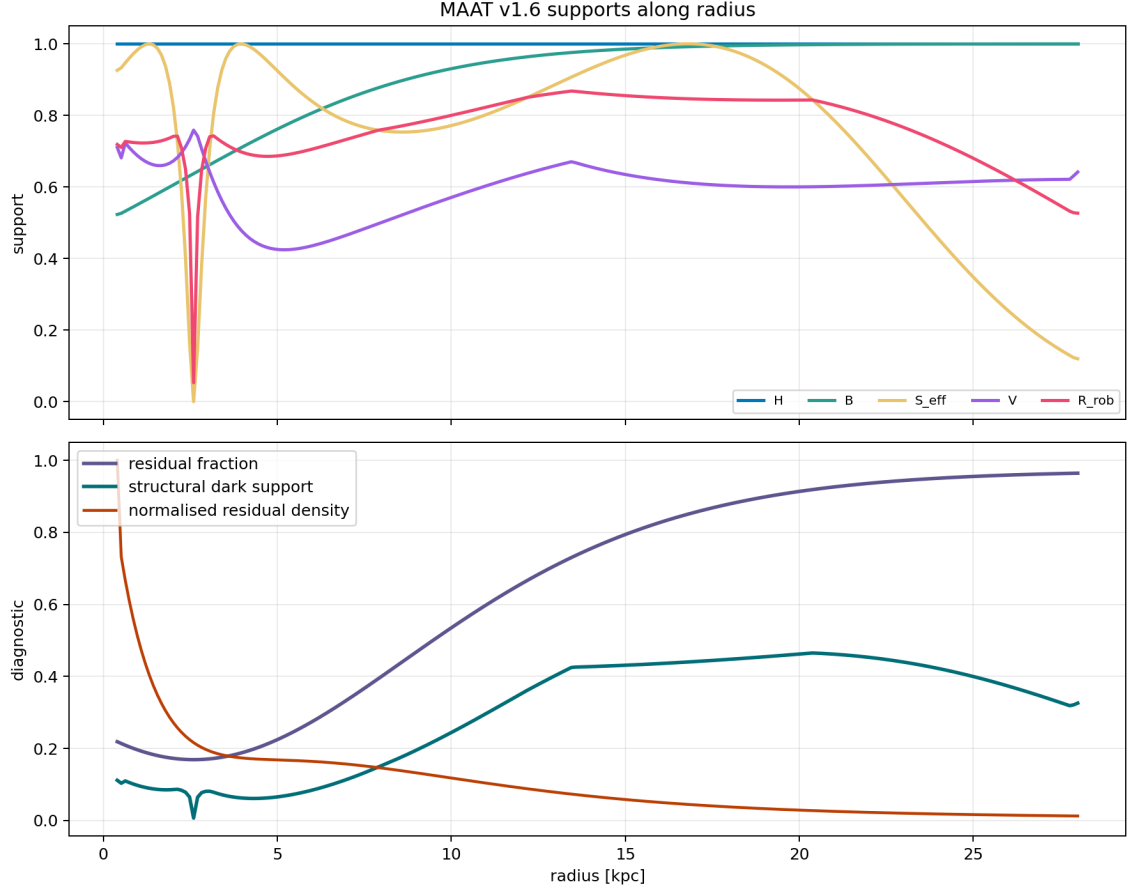


Figure 2: MAAT v1.6 supports along radius. In this toy construction H is near unity because the residual is defined to reconstruct the observed curve. The nontrivial structure is carried by balance, controlled activity, connectedness, and the resulting robustness.

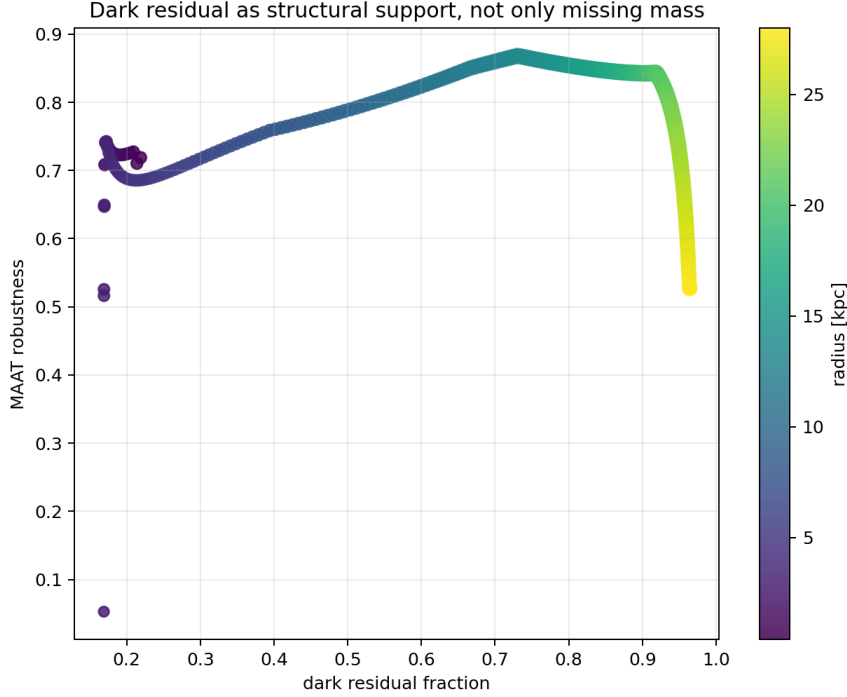


Figure 3: Residual fraction versus robustness. The structural dark support is largest where the residual is dynamically important and remains connected to the coherent profile.

7 Interpretation

In this language, dark matter is not first introduced as a substance. It is first introduced as a closure failure:

$$v_{\text{bar}}^2(r) \not\simeq v_{\text{obs}}^2(r).$$

The effective dark term is the residual that restores the observed dynamical coherence:

$$v_{\text{bar}}^2(r) + v_{\text{dark}}^2(r) \simeq v_{\text{obs}}^2(r).$$

MAAT v1.6 reads this as a search-geometry statement. The visible sector alone does not provide a complete path through the gravitational constraint landscape. The residual channel supplies the missing structural direction.

This gives a plain-language interpretation:

Dark matter is what gravity says must be there so that galaxies remain dynamically coherent.

The sentence is intentionally structural, not microscopic. It does not decide whether the carrier is a particle halo, a field, a modified-gravity response, or a mixture of mechanisms. It says that the observed gravitational dynamics contain a robust residual channel, and any successful microscopic theory must carry that channel without destroying coherence, balance, connectedness, and stability.

8 Beyond Rotation Curves

Real dark matter is not inferred from rotation curves alone. A serious structural interpretation must also survive:

- gravitational lensing and mass maps;
- galaxy clusters and merging systems such as the Bullet Cluster;
- cosmic microwave background constraints;
- large-scale structure formation;
- baryonic feedback and halo-profile degeneracies.

This paper does not address those tests. That limitation is substantial. A MAAT residual that works only for toy rotation curves would have little scientific content. The useful future question is whether the same structural diagnostic can connect rotation residuals, lensing residuals, and cosmological growth residuals without retuning the supports for each case.

9 What Would Make This Nontrivial

The following upgrades would turn the present note into a meaningful test:

- apply the diagnostic to real rotation-curve samples such as SPARC-like galaxy data;
- compare $D_{\text{struct}}(r)$ against standard halo fits, baryonic scaling relations, and MOND-like acceleration relations;
- test whether the support profiles predict residual-shape classes across many galaxies rather than merely describing one synthetic curve;
- include lensing-derived mass profiles and ask whether the same residual support aligns with lensing mass;
- define a cosmological version of the residual and compare it with CMB and large-scale-structure constraints.

Only after such tests would the phrase “structural coherence residual” carry empirical weight beyond being a MAAT-language reformulation of the known dark-matter residual.

10 SPARC/NFW/MOND Upgrade Protocol

The most direct next experiment is a real-galaxy benchmark using the SPARC rotation-curve database of Lelli, McGaugh, and Schombert [5]. SPARC contains mass models and accurate rotation curves for 175 nearby disk galaxies and is widely used as a test bed for galaxy-scale dark-matter and modified-dynamics models.

For each galaxy, the MAAT residual protocol would replace the synthetic profiles by the observed and baryonic SPARC components:

$$v_{\text{bar}}^2(r) = v_{\text{gas}}^2(r) + \Upsilon_{\text{disk}} v_{\text{disk}}^2(r) + \Upsilon_{\text{bulge}} v_{\text{bulge}}^2(r), \quad (8)$$

$$v_{\text{res}}^2(r) = \max[v_{\text{obs}}^2(r) - v_{\text{bar}}^2(r), 0]. \quad (9)$$

The structural score $D_{\text{struct}}(r)$ should then be compared against at least three external baselines.

NFW halo baseline. For a Navarro–Frenk–White halo [6, 7], one may use

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}, \quad (10)$$

with the corresponding halo velocity $v_{\text{NFW}}(r)$ fitted to the same rotation curve. The comparison target is not whether MAAT replaces NFW, but whether $D_{\text{struct}}(r)$ correlates with the fitted residual structure left by NFW-like models across galaxy classes.

MOND/RAR baseline. For a MOND-like or radial-acceleration-relation baseline [8, 9], compare the observed acceleration

$$g_{\text{obs}}(r) = \frac{v_{\text{obs}}^2(r)}{r} \quad (11)$$

to the baryonic acceleration

$$g_{\text{bar}}(r) = \frac{v_{\text{bar}}^2(r)}{r}. \quad (12)$$

The question is whether the MAAT support profiles add information beyond the known baryonic acceleration relation, not whether they merely rediscover it.

Published halo-catalog baseline. As an additional control, one can compare against published SPARC halo catalogues such as Li et al. [10], which fitted multiple halo profiles including NFW, Burkert, Einasto, DC14, coreNFW, and related models. This prevents the MAAT diagnostic from being evaluated only against a weak or single-profile baseline.

Predeclared evaluation. A nontrivial SPARC test should be predeclared before inspecting results:

- Use fixed definitions of $H, B, S_{\text{eff}}, V, R_{\text{rob}}$.
- Report galaxy-by-galaxy and population-level correlations between D_{struct} and residual classes.
- Compare against NFW, MOND/RAR, and published multi-halo fits.
- Use shuffled-radius and shuffled-galaxy nulls to test whether the support structure carries information beyond smooth radial trends.
- Separate high-surface-brightness, low-surface-brightness, dwarf, and bulge-dominated systems.

The falsifying outcome is also clear: if D_{struct} only tracks the same information as v_{res}^2 or g_{bar} and adds no cross-galaxy structure beyond standard baselines, then the MAAT layer is only a renaming of known rotation-curve phenomenology.

11 SPARC Data and License Notes

The current repository does not redistribute SPARC data. All present CSV, JSON, and PNG outputs are synthetic toy artifacts generated by the local script.

If a future version uses SPARC data, the following attribution and licensing rules should be followed:

- Cite the original SPARC paper by Lelli, McGaugh, and Schombert [5].
- If using the Zenodo-hosted SPARC record, cite DOI [10.5281/zenodo.16284118](https://doi.org/10.5281/zenodo.16284118). That record lists the dataset under the Creative Commons Attribution 4.0 International license, so attribution is required.
- If using the VizieR catalogue access route, include the standard CDS acknowledgment: this research has made use of the VizieR catalogue access tool, CDS, Strasbourg, France.
- Do not imply endorsement by the SPARC authors, the SPARC collaboration, Zenodo, VizieR, CDS, or any original data providers.
- Clearly distinguish original SPARC measurements from derived MAAT diagnostic tables and figures.

12 What This Does Not Claim

The present note does not claim:

- to explain the particle identity of dark matter;
- to replace cold dark matter or the standard cosmological model;
- to fit real galaxy rotation curves;
- to address lensing, clusters, CMB constraints, or structure formation;
- to derive a relativistic modified-gravity theory.

The benchmark is narrower: it shows that the galactic dark residual can be operationalised as a structural-coherence diagnostic in MAAT v1.6, while also making clear that this operationalisation is not yet a substantive alternative to established dark-matter phenomenology.

13 Reproducibility

The reproducibility folder is:

```
experiments/dark_matter_structural_coherence/
```

Run:

```
cd experiments/dark_matter_structural_coherence
python3 dark_matter_structural_coherence.py
```

The script writes:

- `dark_matter_structural_coherence_profile.csv`
- `dark_matter_structural_coherence_summary.json`
- `fig1_rotation_curve_residual.png`
- `fig2_structural_supports.png`
- `fig3_residual_robustness_phase.png`

Repository:

<https://github.com/Chris4081/structural-selection-principle>

14 Data Attribution and License Note

No external astronomical dataset is redistributed. The rotation curve, baryonic profile, residual profile, CSV tables, JSON summary, and figures are synthetic reproducibility artifacts generated locally by the script.

Standard dark-matter and rotation-curve literature is cited for scientific context only.

15 Conclusion

MAAT v1.6 does not answer the microscopic identity problem of dark matter. It reframes one familiar part of the phenomenon, the rotation-curve residual, as a structural residual in gravitational search geometry.

The central diagnostic statement is:

dark residual = the missing structural channel required for dynamical coherence.

This is not evidence against particle dark matter, and it is not yet a competitive dark-matter model. Rather, it is a way of expressing what any successful dark-sector theory must accomplish: it must carry the residual gravitational structure while preserving the observed coherence of galaxies and the larger cosmological system. The next meaningful step is to test the diagnostic on real multi-probe data.

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